

UID LAB EXPERIMENT – 10

Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

First, create an HTML file to hold your canvas for the chart and include Chart.js. html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Inventory Management Visualization</title>
<style> body { font-family:
Arial, sans-serif; text-align:
center; margin: 50px;
} canvas { margin:
20px auto;
}

</style>
</head>
<body>
<h1>Inventory Management System</h1>
<canvas id="pieChart" width="400" height="400"></canvas>
```

```
<canvas id="barChart" width="400" height="400"></canvas>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>
</body>
</html>
```

Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic.

javascript script.js

```
//Data for the inventory const inventoryData = { labels:
['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
datasets: [ { label: 'Items in Stock', data: [200, 150, 100, 80, 50],
backgroundColor: [
'#FF6384',
'#36A2EB',
'#FFCE56',
'#4BC0C0',
//
'#9966FF'
],
}
]
};
```

```
//Creating the Pie Chart const ctxPie =
document.getElementById('pieChart').getContext('2d'); const
pieChart = new Chart(ctxPie, {
type: 'pie', data:
inventoryData,
options: {
responsive: true,
```

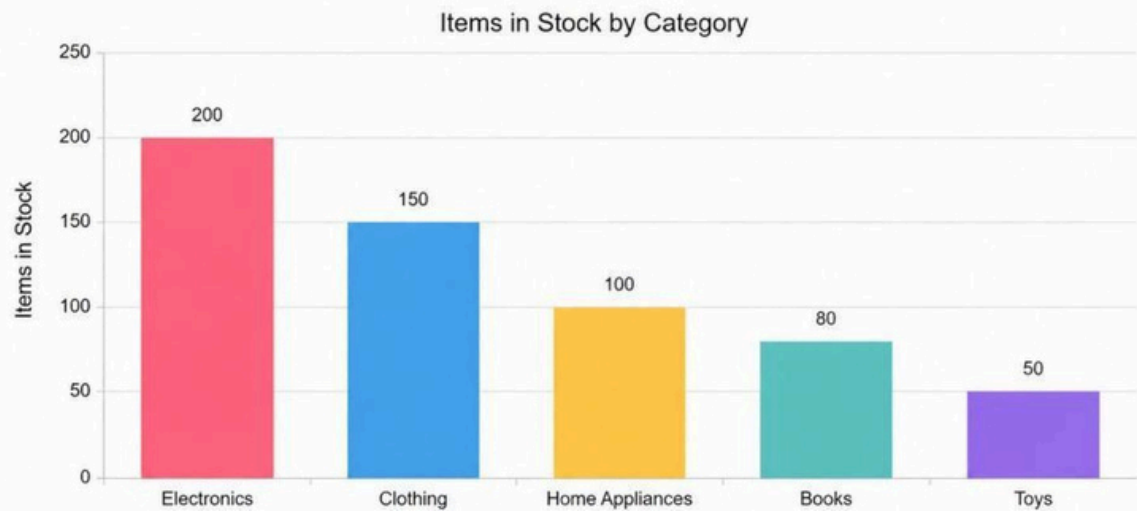
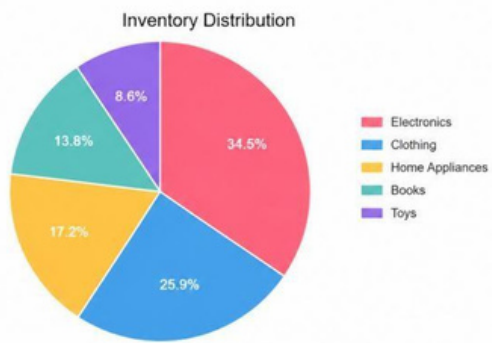
```
title: { display: true, text:
'Inventory Distribution'
}
}
});
```

```
//Creating the Bar Chart const ctxBar =
document.getElementById('barChart').getContext('2d'); const barChart
= new Chart(ctxBar, {
type: 'bar', data:
inventoryData,
options: {
responsive: true,
title: { display:
true,

text: 'Items in Stock by Category'
}, scales: {
yAxes: [{ ticks: {
beginAtZero: true
}

}]
} }
});
```

Inventory Management System



```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Inventory Management Visualization</title>
7 <style>
8 body {
9 font-family: Arial, sans-serif;
10 text-align: center;
11 margin: 50px;
12 }
13 canvas {
14 margin: 20px auto;
15 }
16
17 </style>
18 </head>
19 <body>
20 <h1>Inventory Management System</h1>
21 <canvas id="pieChart" width="400" height="400"></canvas>
22 <canvas id="barChart" width="400" height="400"></canvas>
23 <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
24 <script src="script.js"></script>
25 </body>
26 </html>
27
28 Step 2: Create the JavaScript File for Charts
29 Next, create a JavaScript file (script.js) to handle the data visualization logic.
30 javascript
31 script.js
32
33 // Data for the inventory

```

```

31
32 // Data for the inventory
33 const inventoryData = {
34 labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
35 datasets: [
36 {
37 label: 'Items in Stock',
38 data: [200, 150, 100, 80, 50],
39 backgroundColor: [
40 '#FF6384',
41 '#36A2EB',
42 '#FFCE56',
43 '#4BC0C0',
44 //
45
46 '#9966FF'
47 ],
48 }
49 ]
50 };
51
52 // Creating the Pie Chart
53 const ctxPie = document.getElementById('pieChart').getContext('2d');
54 const pieChart = new Chart(ctxPie, {
55 type: 'pie',
56 data: inventoryData,
57 options: {
58 responsive: true,
59 title: {
60 display: true,
61 text: 'Inventory Distribution'
62 }
63 }
64 });

```